2) V/W_1' has a filtration with successive quotients $W_1, Z_1, \ldots, Z_p$ and another filtration with successive quotients $W_2', \ldots, W_m'$.

By the induction assumption, this means that the collection of irreducible representations with multiplicities $W_1, W_1', Z_1, \ldots, Z_p$ coincides on one hand with $W_1, \ldots, W_n$ and on the other hand with $W_1', \ldots, W_m'$. We are done. $\square$

The Jordan-Hölder theorem shows that the number n of terms in a filtration of V with irreducible successive quotients does not depend on the choice of a filtration and depends only on V. This number is called the **length** of V. It is easy to see that n is also the maximal length of a filtration of V in which all the inclusions are strict.

The sequence of the irreducible representations $W_1, \ldots, W_n$ enumerated in the order they appear from some filtration of V as successive quotients is called a **Jordan-Hölder series** of V.

3.8. The Krull-Schmidt theorem

Theorem 3.8.1 (Krull-Schmidt theorem)**.** *Any finite dimensional representation of A can be uniquely (up to an isomorphism and the order of summands) decomposed into a direct sum of indecomposable representations.*

Proof. It is clear that a decomposition of V into a direct sum of indecomposable representations exists, so we just need to prove uniqueness. We will prove it by induction on $\dim V$. Let $V = V_1 \oplus \cdots \oplus V_m = V_1' \oplus \cdots \oplus V_n'$. Let $i_s : V_s \to V$, $i_s' : V_s' \to V$, $p_s : V \to V_s$, $p_s' : V \to V_s'$ be the natural maps associated with these decompositions. Let $\theta_s = p_1 i_s' p_s' i_1 : V_1 \to V_1$. We have $\sum_{s=1}^n \theta_s = 1$. Now we need the following lemma.

Lemma 3.8.2. *Let W be a finite dimensional indecomposable representation of A. Then:*

(i) Any homomorphism $\theta : W \to W$ is either an isomorphism or nilpotent.

(ii) If $\theta_s : W \to W$, $s = 1, \ldots, n$, are nilpotent homomorphisms, then so is $\theta := \theta_1 + \cdots + \theta_n$.